sample_3-Evaluation report

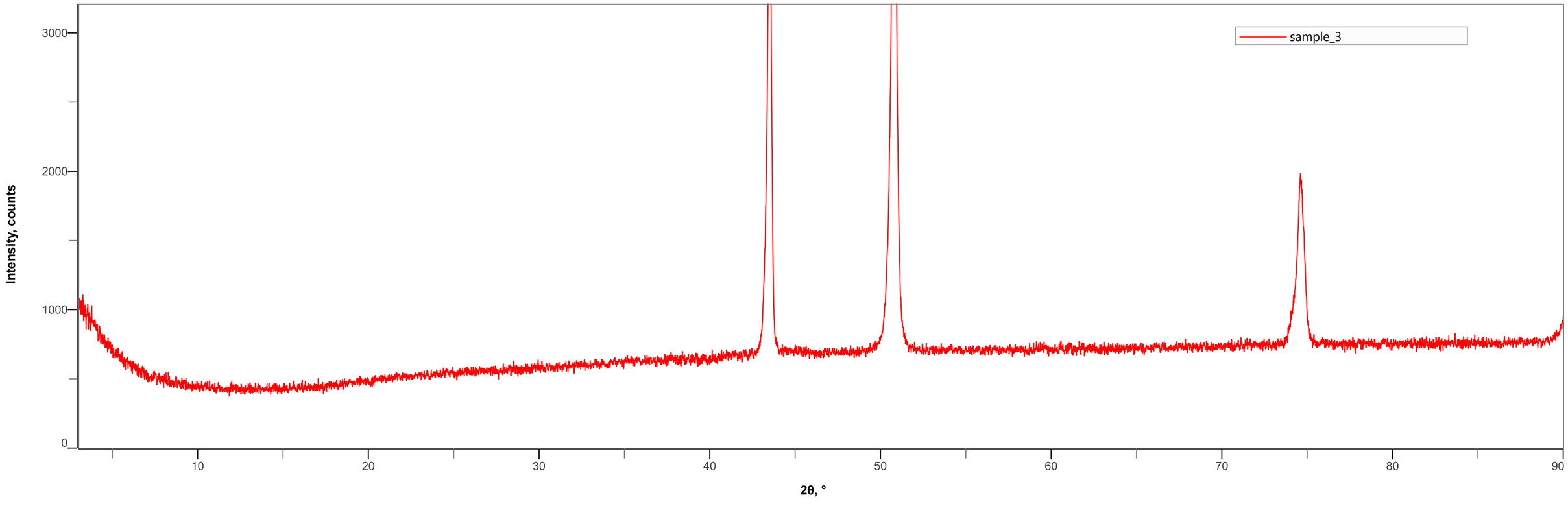
General information

Analysis date	2025-11-27 13:19:17	Measurement start time	2025-11-23 13:01:41
Analyst	Administrator	Operator	Administrator
Sample name	sample_3	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\sample_3.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	Standard sample stage	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

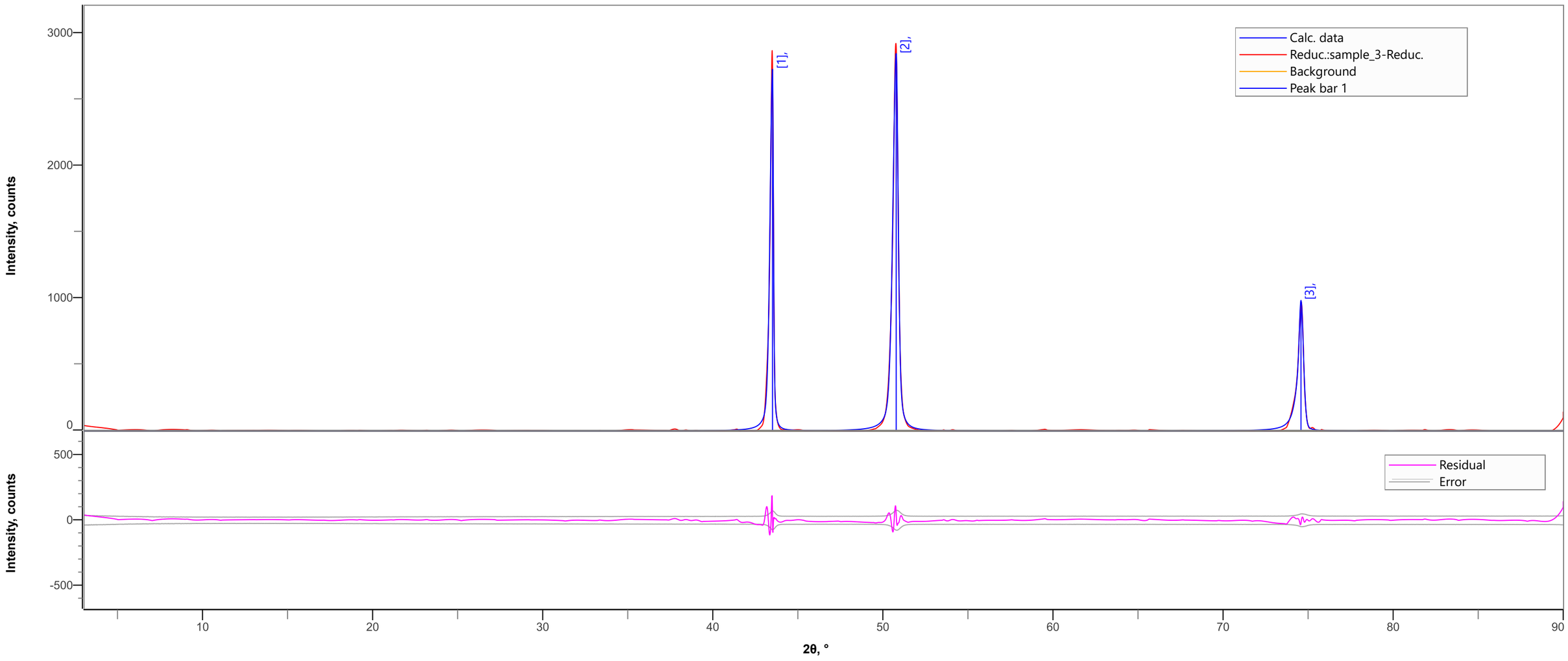
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	25.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

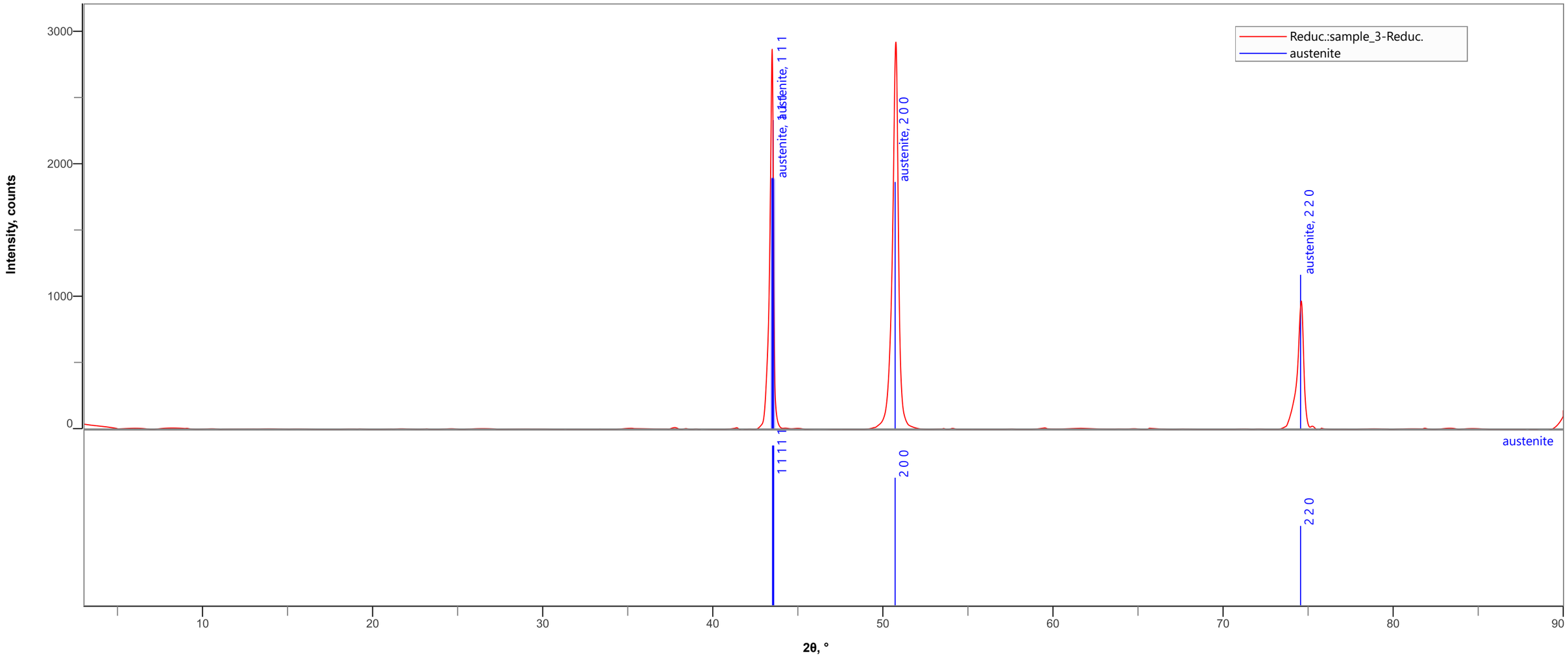
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
austenite	Cr - Ni - Fe - C	0.501	S/M:PDF-5+ 2026	225 : Fm-3m	00-031-0619

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.4873(8)	2.07933(4)	2726(72)	0.2367(8)	829(2)	0.304(9)	2.64(5)	0.483(8)	0.66(2)	377.4(13)	austenite: 1 1 1	Cr - Ni - Fe - C	00-031-0619	62.24	Split Pseudo-Voigt	-	-	-	-	-	
2	50.7542(11)	1.79735(3)	2843(74)	0.3661(10)	1332(3)	0.469(13)	1.61(2)	0.485(7)	0.580(12)	250.8(7)	austenite: 2 0 0	Cr - Ni - Fe - C	00-031-0619	100.00	Split Pseudo-Voigt	-	-	-	-	-	
3	74.538(2)	1.27204(3)	983(50)	0.349(3)	503(2)	0.51(3)	1.04(3)	1.22(2)	0.29(3)	299(2)	austenite: 2 2 0	Cr - Ni - Fe - C	00-031-0619	37.78	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
austenite	3.59879	3.59879	3.59879	90.000	90.000	90.000

d-I List

austenite

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.52169	2.07777	1 1 1	100.00
2	50.69229	1.79940	2 0 0	80.00
3	74.51595	1.27237	2 2 0	50.00

Simple Inverse Pole Figure

austenite

